

SpaceX Falcon 9 First-Stage Landing Prediction

Data Science Capstone Project Report

GitHub repository: <https://github.com/razerblade09/spacex-falcon9-capstone>

Executive Summary

Methods used: Data was collected via the SpaceX REST API and Wikipedia web scraping, cleaned and labeled, explored with visualization and SQL, mapped geospatially with Folium, and modeled with four classifiers (Logistic Regression, SVM, Decision Tree, KNN) tuned via GridSearchCV.

Key results: Across 90 Falcon 9 launches, the overall first-stage landing success rate was 67%. The best-performing model, Logistic Regression, reached 83% test accuracy in predicting landing outcome.

90

Launches analyzed

67%

Overall landing success rate

83.3%

Best model test accuracy

4

ML models compared

Conclusion: Landing success is highly predictable from launch parameters, meaning a competing launch provider could reliably estimate SpaceX's true reusable-launch cost advantage — see full Conclusion & Insights on the final slide.

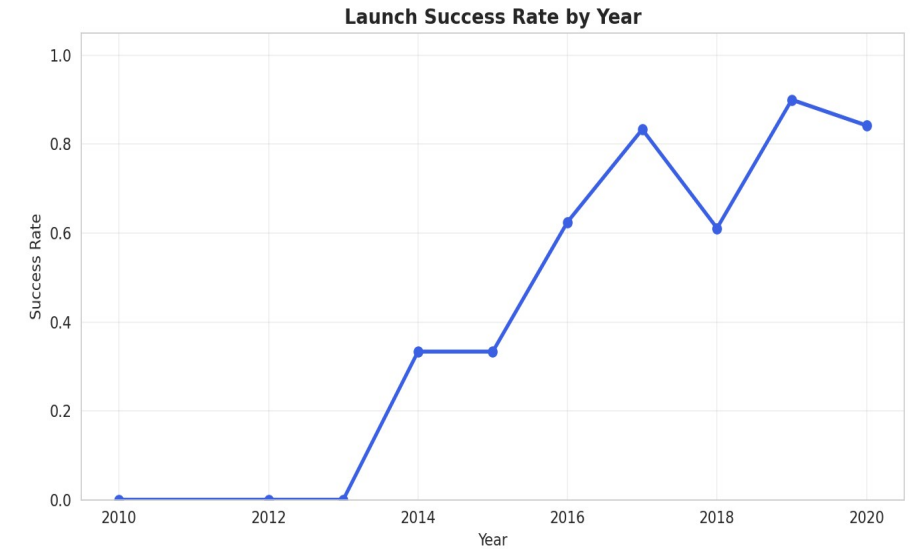
Introduction

SpaceX advertises Falcon 9 launches at \$62 million, versus \$165M+ from other providers — largely because SpaceX reuses the rocket's first stage instead of discarding it.

Problem statement: If the first stage does not land successfully, that launch's cost estimate changes substantially. This project builds a model to predict whether the first stage will land, using historical launch data — enabling accurate cost estimation for a competing bid.

Objectives:

- Collect and clean historical Falcon 9 launch data
- Identify factors correlated with landing success
- Visualize patterns across sites, orbits, and time
- Build and evaluate a landing-outcome classifier



Falcon 9 landing success climbed steadily as SpaceX matured its recovery process.

SpaceX API & Web Scraping

SpaceX REST API

GET /v4/launches/past for all historical launches, filtered to single-core Falcon 9 flights, then enriched via /rockets, /launchpads, /payloads, /cores to resolve IDs into readable attributes.

Result: 90 clean Falcon 9 launch records (2010–Nov 2020): booster version, payload mass, orbit, site, grid fins, reused flag, legs, landing pad, outcome.

Falcon 1 launches excluded — this project targets Falcon 9 landing prediction only.

Wikipedia Web Scraping

Requested the "List of Falcon 9 and Falcon Heavy launches" page, parsed HTML wikitables with BeautifulSoup, and cleaned each row's text (dates, booster serials, payload mass) into a dataframe.

Purpose: an independent, cross-checked source used to validate the API dataset and fill fields (like precise payload customer) the API doesn't expose.

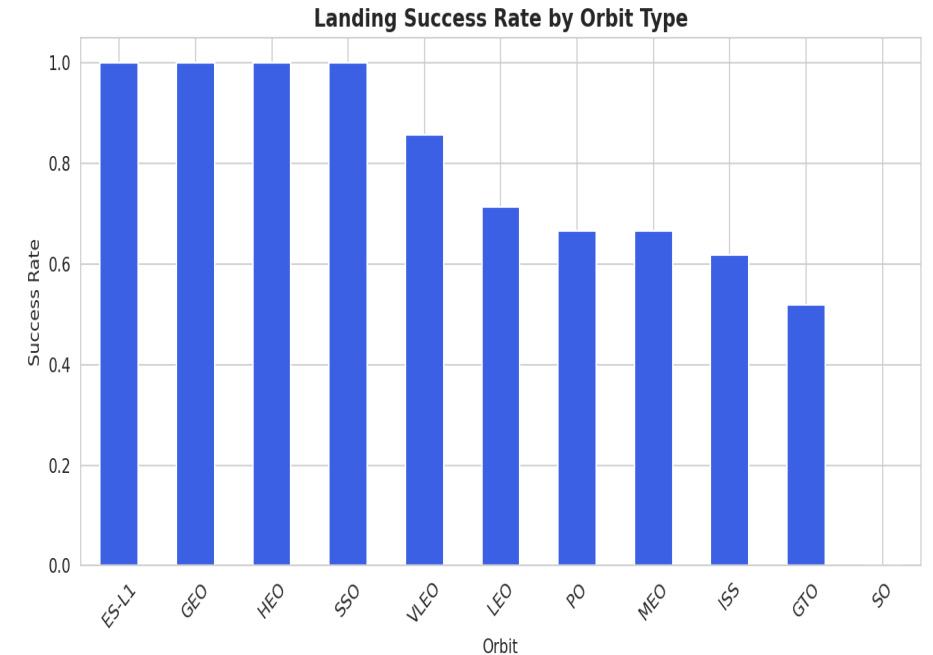
Key techniques: regex/Unicode cleanup for payload-mass strings; helper functions to pull booster version and landing status out of merged table cells.

Data Wrangling Methodology

Missing values: PayloadMass (5.6% missing) filled with column mean; LandingPad (28.9% missing) kept as-is since a null pad legitimately means "no ground/drone-ship landing occurred."

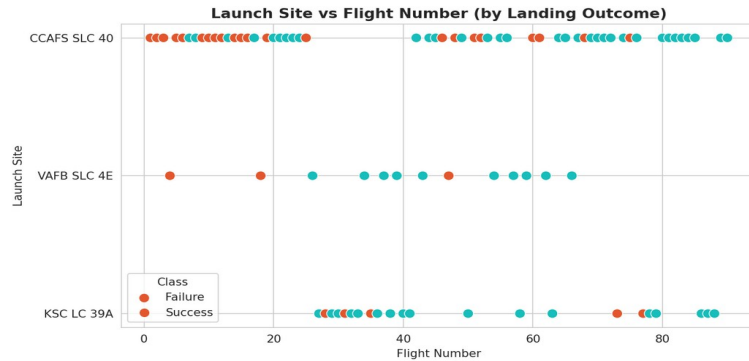
Label engineering: The raw Outcome field (e.g. "True ASDS", "False Ocean", "None None") was converted into a binary Class label — 1 if the booster landed successfully by any method, 0 otherwise.

Result: 90 launches → 60 successful landings (66.7%), 30 unsuccessful (33.3%)

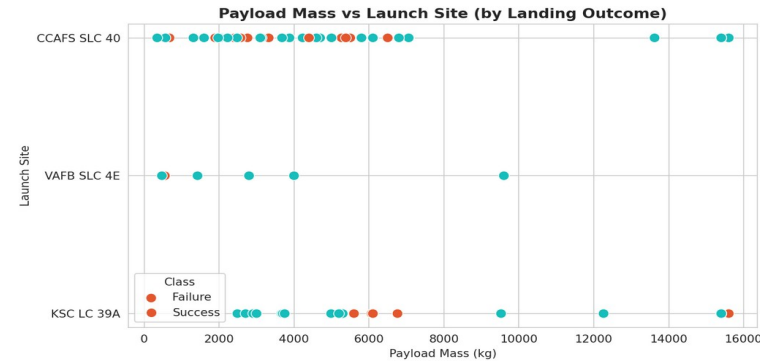


EDA with Data Visualization

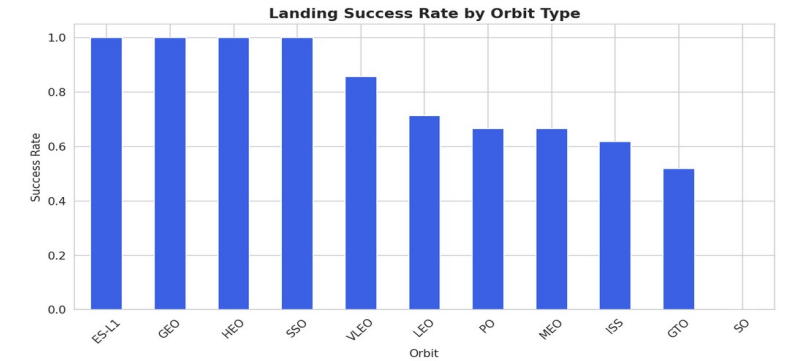
We plotted flight number, payload mass, orbit type, and launch year against landing outcome to surface which factors correlate with success.



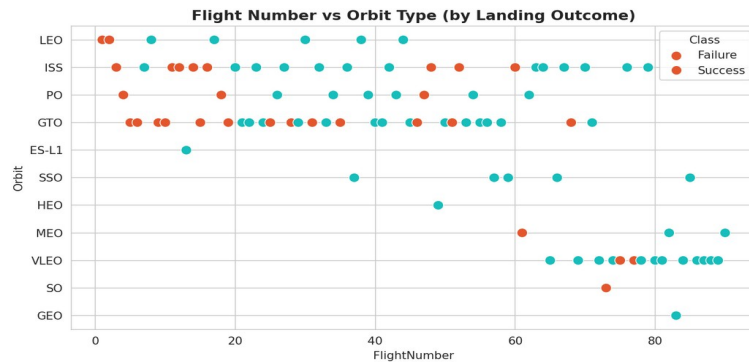
Flight # vs Site



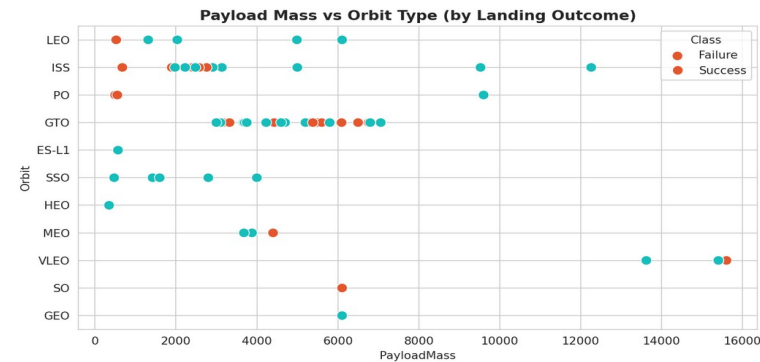
Payload vs Site



Success Rate by Orbit



Flight # vs Orbit



Payload vs Orbit



Success Rate by Year

Takeaway: GTO/ISS-orbit missions (heavier, higher-energy) show lower success rates; success climbed year-over-year, from ~0% in 2010–2013 to near 100% by 2019–2020. Chart types used: scatter plots and bar/line charts.

EDA with SQL

The scraped launch table was loaded into a SQL database (SQLite / IBM Db2) as SPACEXTABLE and queried to answer specific business questions:

- ✓

Distinct launch sites used
- ✓

Total payload mass for NASA (CRS)
- ✓

Avg. payload mass, F9 v1.1 boosters
- ✓

Date of first successful ground-pad landing
- ✓

Boosters landing on drone ship, 4–6t payload
- ✓

Count of each mission outcome
- ✓

Booster(s) carrying max payload mass
- ✓

Failed drone-ship landings in 2015
- ✓

Ranking of landing outcomes 2010–2017

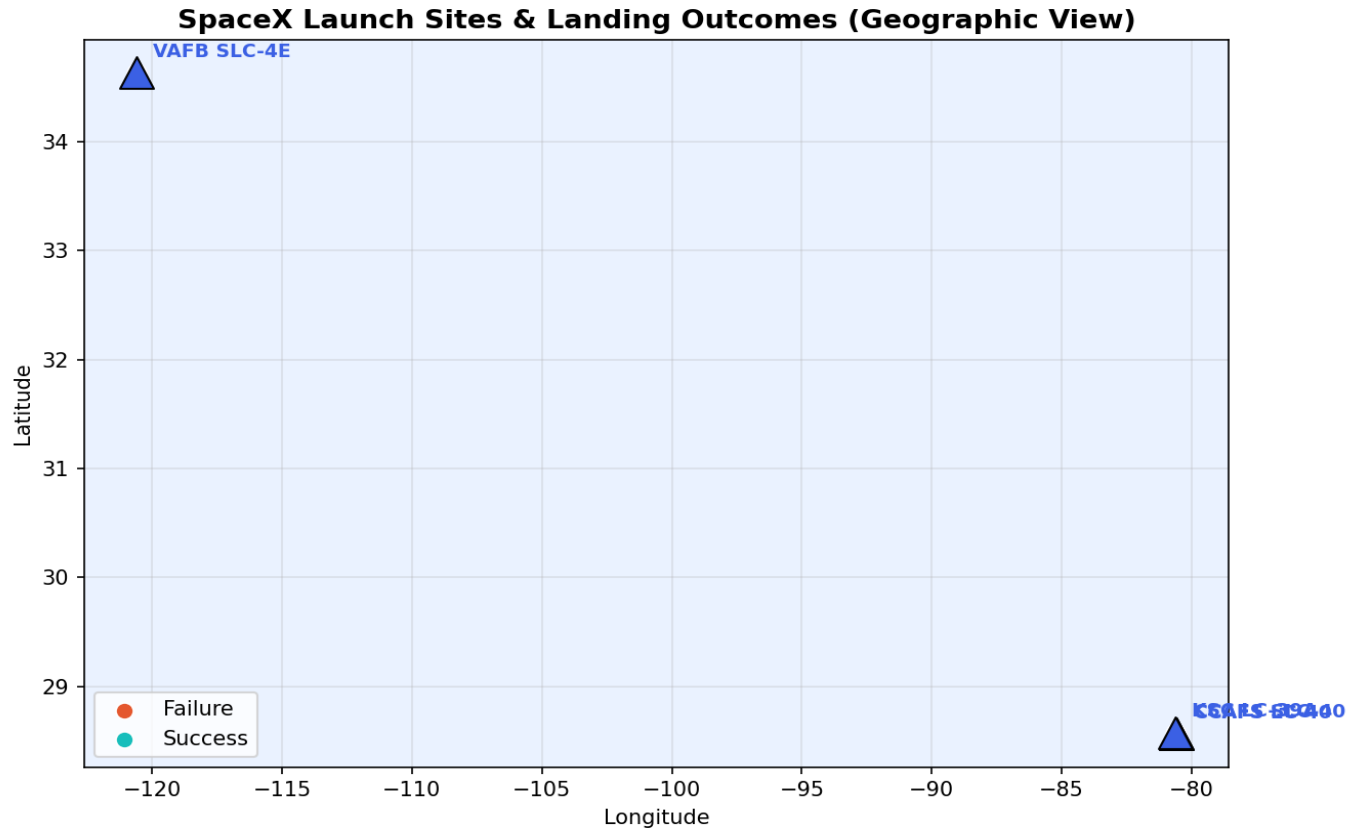
Query Results

4 launch sites: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
NASA (CRS) total payload: 45,596 kg
Max single payload: 15,600 kg (F9 B5)
First successful ground-pad landing: 1 May 2017
Mission outcomes 2010–2020: 98 of 101 successful (booster-recovery success is the separate, harder metric this model targets)

Landing Outcome Ranking (2010–2017, top 5 by count)

Outcome	Count
Success	20
No attempt	10
Success (drone ship)	8
Success (ground pad)	6
Failure (drone ship)	4

Folium Map

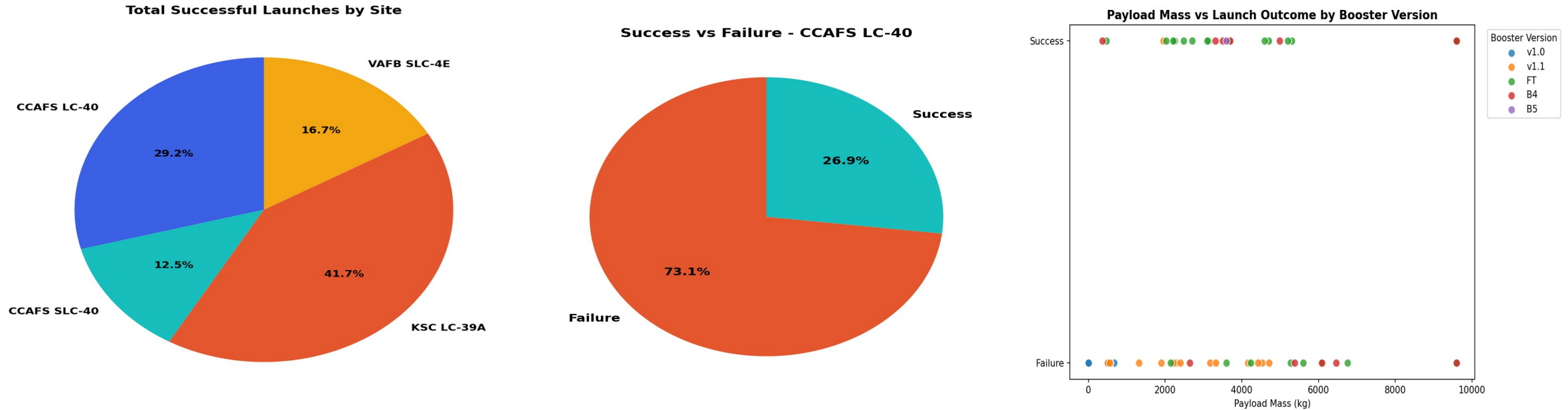


All 4 launch sites plotted with markers. Every individual launch (55 records) is a marker, colored green for a successful landing and red for a failed/no-attempt landing — showing which sites cluster more successes.

Observation: KSC LC-39A and CCAFS SLC-40 show visibly denser green clusters than VAFB SLC-4E, consistent with higher launch cadence and a more refined recovery process.

Proximity analysis (haversine distance from KSC LC-39A): nearest coastline point 7.43 km; nearest city (Titusville area) 20.50 km. Sites are deliberately close to coastlines for downrange safety but a safe distance from population centers — a siting constraint any competing provider must replicate.

Plotly Dash App



Dropdown to filter by launch site drives a live-updating success-rate pie chart (left: share of successful landings per site; center: success/failure split at the top-volume site, CCAFS LC-40) and a **RangeSlider (0–10,000 kg)** narrows the payload window on the scatter plot (right), which recolors by booster version (v1.0/v1.1/FT/B4/B5).

Insight: newer booster generations succeed reliably across a much wider payload range than early versions — reusability maturity, not just payload size, drives success.

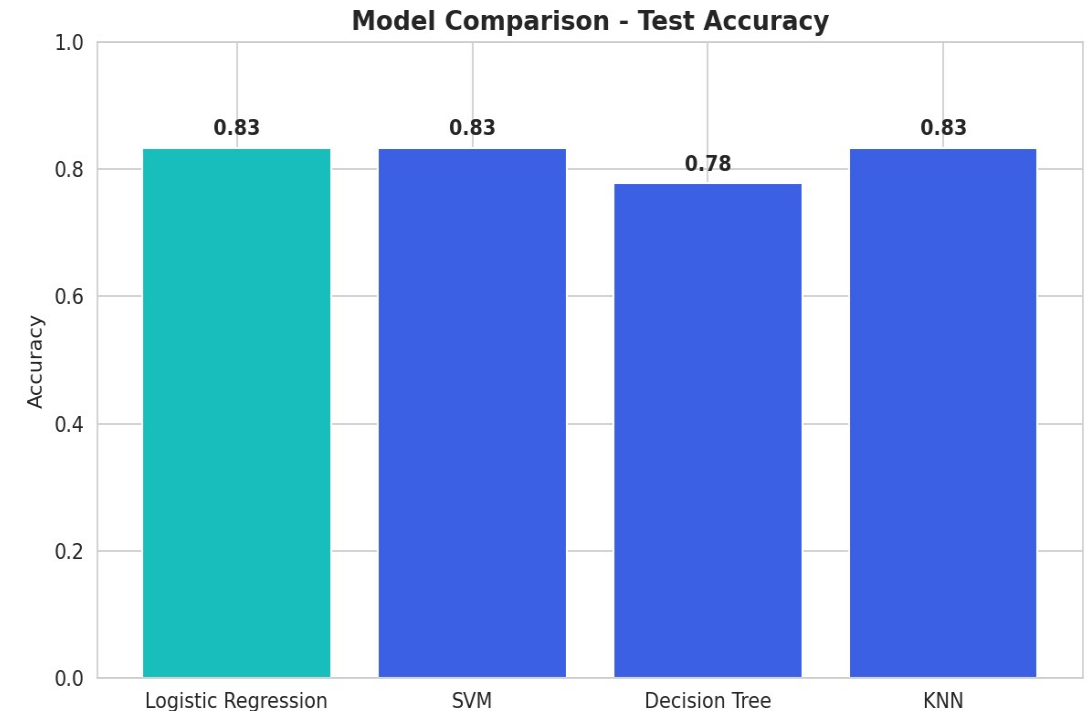
Model Development & Comparison

Features: 80 one-hot-encoded columns (orbit, launch site, landing pad, booster serial) plus flight number, payload mass, flights, grid fins, reused, legs, block — standardized with StandardScaler.

Split: 80% train / 20% test (72 / 18 launches), stratified, random_state=2.

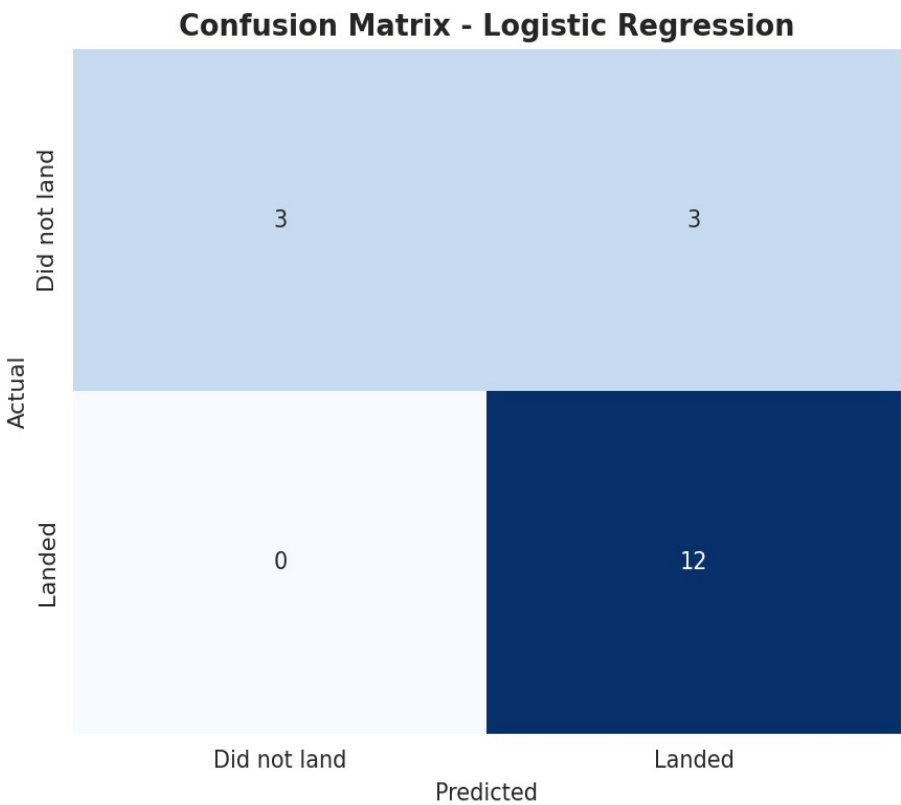
Models compared (10-fold GridSearchCV each):

- Logistic Regression — C, penalty, solver
- Support Vector Machine — kernel, C, gamma
- Decision Tree — criterion, max_depth, min_samples_split
- K-Nearest Neighbors — n_neighbors, distance metric (p)



All four models converged near 83% test accuracy, with Decision Tree slightly behind — the engineered features are strong, and model choice matters less than feature quality here.

Evaluation Results & Best Model



Model	CV Score	Test Acc.
Logistic Regression	0.821	0.833
SVM	0.834	0.833
KNN	0.834	0.833
Decision Tree	0.804	0.778

Confusion matrix (18 test launches): 12 successful landings correctly predicted, 3 unsuccessful correctly predicted, 3 unsuccessful misclassified as successful, 0 successful landings missed — the model is conservative in the safer direction for cost bidding.

Best model: Logistic Regression (C=1, L2 penalty, lbfgs solver). Ties SVM/KNN on accuracy (83.3%) but is far more interpretable — each coefficient directly shows its effect on landing-success odds, which matters for explaining a cost bid to stakeholders, and it's the cheapest to retrain as new launches arrive.

Conclusion & Insights

1

Landing success is highly learnable from launch parameters — 83% test accuracy from a simple, interpretable model (Logistic Regression).

2

Orbit type and payload mass are the strongest predictors; heavier GTO missions land less reliably than light LEO/ISS missions.

3

Success rate improved dramatically over time (2013 → 2020), reflecting SpaceX's iterative process improvements — not a static property of the hardware.

4

Interactive tools (Folium map, Plotly Dash) let stakeholders verify these patterns themselves rather than trusting static charts alone.

5

Recommendation: a competing bidder can use this model as a first-pass cost-risk estimator, refining it further with more recent launches and continuous re-training.